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Drop Queen

TECHNICAL ARCHITECTURE — CISC 610 MLOPS · SPRING 2026

"GoViralIQ predicts what goes viral. Drop Queen predicts what sells because of it."

01 · DATA SOURCES

🗄️ Raw Data Ingestion

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Amazon Sales Data

Historical product demand, sales rank, and review data across beauty & lifestyle categories

KAGGLE

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TikTok Trend Data

Viral product engagement signals — views, shares, hashtag volume, and shop clicks

KAGGLE API

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Google Trends API

Real-time search interest signals by product keyword and beauty category

FREE API

02 · DATA PIPELINE

Processing & Feature Engineering

Data Cleaning

Remove nulls, duplicates, and outliers from raw datasets

PANDAS

Feature Engineering

Create lag features, rolling averages, and social signal scores

NUMPY

Data Merging

Join Amazon sales data with TikTok signals by product & date

PANDAS

EDA & Validation

Exploratory analysis and data quality checks before modeling

SCIKIT-LEARN

03 · MACHINE LEARNING MODELS

Dual Model Architecture

Model 1 — Time Series Forecasting

Predicts weekly product demand 2 weeks in advance using historical Amazon sales patterns. Trained with Prophet and LSTM neural networks.

Prophet LSTM TensorFlow MLflow

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Model 2 — Trend Detection

Detects which products are gaining social momentum before they peak. Uses TikTok engagement signals and Google Trends as early warning indicators.

XGBoost Scikit-learn Feature Store MLflow

👑 Ensemble Layer

Both models are combined into a hybrid ensemble that outputs a unified demand spike score + confidence rating per product

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04 · SERVING LAYER

⚡ Flask REST API — Dockerized & Deployed

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POST /predict/demand

Returns weekly demand forecast for a given product ID and platform

FLASK

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POST /predict/trend

Returns trend detection score and predicted spike timing for a product

FLASK

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GET /products/top

Returns top 10 predicted product drops for the coming week

FLASK

🐳 Docker Container | 🔒 REST API | 📋 Pylint | 🧪 Pytest | 📦 requirements.txt

05 · AUTOMATION & OUTPUT

🔄 CI/CD Pipeline + Dashboard

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GitHub Actions CI/CD

On every push to main: runs Pylint → Pytest → builds Docker image → retrains models on new data → registers new version in MLflow → deploys updated API automatically

GitHub Actions | MLflow Registry | Auto Retrain

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Demo Dashboard

Live web interface showing predicted product drops, demand spike charts, trend scores, and cross-platform TikTok Shop vs Amazon comparisons in real time

Flask Templates | Charts | Live Predictions

👑 Your Portfolio Story

Project 01

Melanin Match AI

Computer Vision · Skin tone matching for beauty products

Project 02

GoViralIQ

ML · Predicts what content goes viral on social media

Project 03 · Current

Drop Queen 👑

MLOps · Predicts which products sell because of viral content

"I understand the full journey — from viral content creation → to product demand → to purchase behavior"
